

GIT-HOL 3

Branching and Merging

Objective

- Create a branch, make changes, merge with main/master.

Prerequisites

- Completed HOL 2.

Procedure

1. Create GitNewBranch.
2. Switch to the branch.
3. Create branch.txt.
4. Stage and commit changes.
5. Switch back to main.
6. Compare branches.
7. Merge GitNewBranch into main.
8. View log and delete merged branch.

Commands

```
git branch GitNewBranch
git checkout GitNewBranch
echo "This file is created in GitNewBranch" > branch.txt
git add branch.txt
git commit -m "Added branch.txt"
git checkout main
git diff main GitNewBranch
git merge GitNewBranch
git log --oneline --graph --decorate
git branch -d GitNewBranch
git push origin main
```

Output

```
MINGW64~/c/Users/lahar/GitDemo
lahar@laharitata MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git branch GitNewBranch

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git branch
* GitNewBranch
* main

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

lahar@laharitata MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file is created in GitNewBranch" > branch.txt

lahar@laharitata MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

lahar@laharitata MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch 84b5bea] Added branch.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branch.txt

lahar@laharitata MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

lahar@laharitata MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git diff main GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..a151d9d
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git merge GitNewBranch
Updating 67dfc49..84b5bea
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
create mode 100644 branch.txt
```

```
MINGW64~/c/Users/lahar/GitDemo
lahar@laharitata MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
* 84b5bea (HEAD -> main, GitNewBranch) Added branch.txt in GitNewBranch
* 67dfc49 (origin/main) Added gitignore file
* 5e3d4eb Added welcome file

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 84b5bea).

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 365 bytes | 365.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/laharitata/GitDemo.git
67dfc49..84b5bea main -> main
```

Result

Branch was successfully merged into the main branch.